GROUP 16

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THE PHARMACY INVENTORY AND SALES MANAGE MENT SYSTEM

1. **PROBLEM**:
   Modern pharmacy struggle with operational vulnerabilities that include inventory vulnerabilities that include mismanagement of inventory, financial discrepancies and financial losses resulting from expired medicines and without structured tracking, pharmacies face challenges in making sure that drugs with shorter expiry dates are distributed first.

   Additionally, they face a problem of manually and poorly record-keeping which creates risks regarding staff accountability and intake of stock.

2. **SOLUTION**
   The PharmaFlow DBMS is a robust database management system that is tailored to automate and secure core pharmacy workflows, providing an integrated digital infrastructure for categorized drug catalogs, supplier tracking, inventory tracking, sales processing and role-based operational control.

3. **OBJECTIVES**
   - To establish operational security by implementing four primary user roles.
   - To track stock-level variations, batches, manufacturing dates and expiry dates for drugs.
   - To maintain complete financial accountability by tying every transaction to a specific staff member or user.
   - To prevent stock losses by enforcing the prioritization of drugs nearing their expiry dates.
   - To organize pharmaceutical stock logically using categorial classifications and maintain clean supplier vendor logs for seamless restocking.

4. LOGIC AND FUNCTIONALITY

- FIFO Expiry Management: Stock with short expiry dates is let out first to prevent expired stock losses.

- Role-Based Access Control: The pharmacy manager handles users, business analytics and high-level reports. The inventory manager handles deliveries and batches. The pharmacists validate prescriptions and manage dispensing. The cashier processes cash and mobile money payments.

- Staff accountability: This ensures that multiple checkout transactions over time are each tied to the user who worked on them via foreign keys of their USERID's.

- Sales-tracking: Since every transaction can contain multiple purchased drugs, the system instantly deducts quantities from the correct inventory batch when a sale is made.